Employee Attrition Data Analysis Project

Professional Data Analysis Project Document

Project Title: Employee Attrition Analysis Using Python
Subtitle: Exploratory Data Analysis of Employee Characteristics and Attrition
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Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab/Jupyter Notebook
Dataset: `WA_Fn-UseC_-HR-Employee-Attrition.csv`

1. Cover Page

Employee Attrition Analysis Using Python

This project analyzes employee information to understand the structure of an HR dataset and explore employee attrition-related patterns.

The analysis was performed using Python libraries including **Pandas, NumPy, Matplotlib, and Seaborn**.

The uploaded notebook loads the HR Employee Attrition dataset and performs data loading, exploration, missing-value checking, statistical analysis, visualization, and correlation analysis. The dataset contains **1,470 employee records and 35 columns**.

2. Executive Summary

Employee attrition means employees leaving an organization. Understanding employee attrition is important because employee turnover can affect recruitment costs, productivity, team stability, and organizational performance.

This project uses an HR Employee Attrition dataset containing **1,470 rows and 35 columns**. The dataset contains employee-related information such as age, department, business travel, distance from home, education, job role, job satisfaction, monthly income, overtime, performance rating, total working years, work-life balance, years at company, and attrition status.

The Python notebook begins by importing Pandas, NumPy, Seaborn, and Matplotlib. The dataset is loaded into a Pandas DataFrame and inspected using functions such as `head()`, `tail()`, `info()`, `describe()`, `shape`, `columns`, and `dtypes`.

The dataset contains **26 integer columns and 9 object/text columns**. All 1,470 records have non-null values according to the `info()` output. The notebook also checks missing values using `isnull()` and `isnull().sum()`, with zero missing values reported for every column.

The exploratory analysis includes an age distribution histogram, a scatter plot comparing years at the company with monthly income, a scatter plot showing the same relationship by attrition status, and a correlation heatmap for numerical variables.

The analysis provides a foundation for understanding employee characteristics and identifying relationships that may be useful for HR decision-making. However, the uploaded notebook does **not** contain enough completed grouped calculations to make unsupported claims such as which department has the highest attrition or which job role has the highest turnover.

The final outcome is a structured exploratory analysis that can be extended into an HR dashboard, predictive attrition model, or employee-retention project.

3. Introduction

3.1 Project Overview

This project focuses on analyzing an employee HR dataset using Python.

The main subject of the dataset is **employee attrition**. The `Attrition` column contains values such as `Yes` and `No`, representing whether an employee left the organization.

The dataset also contains information about:

- Employee age
- Department
- Job role
- Business travel
- Education
- Job satisfaction
- Monthly income
- Overtime
- Work-life balance
- Years at the company
- Total working experience
- Performance rating
- Relationship satisfaction

3.2 Importance of the Analysis

Employee attrition analysis can help organizations understand employee characteristics and investigate factors associated with employee turnover.

An HR team can use such analysis to:

- Monitor employee turnover
- Understand workforce characteristics
- Identify areas requiring further investigation
- Support employee-retention strategies
- Improve HR reporting

3.3 Expected Outcome

The expected outcome is a clean understanding of the dataset, supported by statistical exploration and meaningful visualizations.

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4. Problem Statement

Organizations need to understand employee attrition because employee turnover can create operational and financial challenges.

The HR dataset provides information about employees and their attrition status. The problem is to explore this information and identify useful patterns and relationships that can support HR decision-making.

The analysis focuses on questions such as:

- What does the employee population look like?
- What are the main employee characteristics?
- How are employee age and experience distributed?
- How does monthly income relate to years at the company?
- How do numerical variables relate to each other?
- Which variables should be investigated further in relation to attrition?

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5. Project Objectives

The main objectives are:

1. Understand the HR dataset.
2. Load the dataset using Pandas.
3. Examine the number of rows and columns.
4. Understand column data types.
5. Check for missing values.
6. Perform descriptive statistical analysis.
7. Explore employee age distribution.
8. Analyze the relationship between years at company and monthly income.
9. Compare the income/experience relationship by attrition status.
10. Examine correlations among numerical variables.

11. Create meaningful visualizations.
12. Generate data-supported HR insights.
13. Identify limitations in the current analysis.
14. Suggest future improvements.

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6. Dataset Description

6.1 Dataset Name

HR Employee Attrition Dataset

The notebook loads:

`WA_Fn-UseC_-HR-Employee-Attrition.csv`

The dataset contains:

Property	Value
Rows	1,470
Columns	35
Numerical columns	26
Object/Text columns	9
Date columns	0 reported
Missing values	0 reported

The `shape` output is `(1470, 35)`, and `info()` reports 1,470 non-null records for every column.

6.2 Column Description

The following descriptions are based on the column names used in the notebook. The uploaded notebook does not provide a formal data dictionary, so these descriptions should be treated as practical interpretations of the field names.

Column	Type	Description
Age	Numerical	Employee age
Attrition	Categorical	Whether the employee left

Column	Type	Description
BusinessTravel	Categorical	Employee business travel frequency
DailyRate	Numerical	Daily rate value
Department	Categorical	Employee department
DistanceFromHome	Numerical	Distance between home and workplace
Education	Numerical/Categorical Code	Education level code
EducationField	Categorical	Employee education field
EmployeeCount	Numerical	Employee count field
EmployeeNumber	Numerical	Employee identification number
EnvironmentSatisfaction	Numerical/Categorical Code	Workplace environment satisfaction rating
Gender	Categorical	Employee gender
HourlyRate	Numerical	Hourly rate
JobInvolvement	Numerical/Categorical Code	Job involvement rating
JobLevel	Numerical/Categorical Code	Employee job level
JobRole	Categorical	Employee job role
JobSatisfaction	Numerical/Categorical Code	Job satisfaction rating
MaritalStatus	Categorical	Employee marital status
MonthlyIncome	Numerical	Monthly income
MonthlyRate	Numerical	Monthly rate
NumCompaniesWorked	Numerical	Number of companies previously worked for
Over18	Categorical	Whether employee is over 18
OverTime	Categorical	Whether employee works overtime
PercentSalaryHike	Numerical	Percentage salary increase

Column	Type	Description
PerformanceRating	Numerical/Categorical Code	Performance rating
RelationshipSatisfaction	Numerical/Categorical Code	Relationship satisfaction rating
StandardHours	Numerical	Standard working hours
StockOptionLevel	Numerical/Categorical Code	Stock option level
TotalWorkingYears	Numerical	Total years of work experience
TrainingTimesLastYear	Numerical	Training sessions/times during previous year
WorkLifeBalance	Numerical/Categorical Code	Work-life balance rating
YearsAtCompany	Numerical	Years spent at the company
YearsInCurrentRole	Numerical	Years in current role
YearsSinceLastPromotion	Numerical	Years since last promotion
YearsWithCurrManager	Numerical	Years working with current manager

The notebook's `columns` output confirms these 35 field names.

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7. Business Questions

The following questions can be investigated using the available fields:

1. What is the age distribution of employees?
2. What is the relationship between years at the company and monthly income?
3. Does the relationship between years at the company and monthly income differ by attrition status?
4. How are employee satisfaction variables distributed?
5. How are employees distributed across departments?
6. How are employees distributed across job roles?

7. How is overtime related to attrition?
8. How is business travel related to attrition?
9. How does distance from home relate to employee attrition?
10. What numerical variables show strong correlations?
11. How does total working experience relate to monthly income?
12. How does job level relate to monthly income?

Important: The current notebook does not contain completed calculations for every question above. These are recommended business questions for further analysis rather than claimed findings.

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8. Tools and Technologies

Tool	Purpose	Real-World Use
Python	Main programming language	Data analysis and automation
Pandas	Data manipulation	Cleaning and analyzing tabular data
NumPy	Numerical operations	Mathematical and statistical calculations
Matplotlib	Visualization	Creating analytical charts
Seaborn	Statistical visualization	Creating attractive statistical plots
Google Colab	Development environment	Running Python notebooks online
Jupyter Notebook	Interactive analysis	Data science projects and experiments
VS Code	Code editor	Developing Python projects

The uploaded notebook imports Pandas, NumPy, Seaborn, and Matplotlib.

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9. Project Workflow

```
Problem Statement
        ↓
Dataset Collection
        ↓
Import Libraries
        ↓
Load Dataset
        ↓
Data Exploration
        ↓
Data Cleaning
        ↓
Exploratory Data Analysis
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Univariate Analysis
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Bivariate Analysis
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Multivariate Analysis
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Feature Engineering
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Data Visualization
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Business Insights
        ↓
Recommendations
        ↓
Conclusion
```

Workflow Explanation

1. Problem Statement

Define the HR problem and determine what the analysis should investigate.

2. Dataset Collection

Load the employee attrition dataset.

3. Import Libraries

Import the Python libraries required for analysis.

4. Load Dataset

Read the CSV file into a Pandas DataFrame.

5. Data Exploration

Understand the size, columns, data types, and statistical characteristics.

6. Data Cleaning

Check missing values and other potential data-quality problems.

7. EDA

Explore employee characteristics and relationships between variables.

8. Visualization

Convert important patterns into charts.

9. Business Insights

Translate statistical observations into understandable HR findings.

10. Recommendations

Suggest areas for further investigation or business action.

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10. Data Loading

The notebook imports four major libraries:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

The dataset is loaded using:

```
df = pd.read_csv("/content/WA_Fn-UseC_-HR-Employee-Attrition.csv")
```

What is a DataFrame?

A **DataFrame** is a table-like structure in Pandas.

It contains:

- Rows representing records
- Columns representing variables

In this project:

- Each row represents an employee record.
- Each column represents an employee-related attribute.

The dataset was successfully loaded and displayed with 1,470 rows and 35 columns.

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11. Data Exploration

11.1 `head()`

Syntax

```
df.head()
```

Purpose

Displays the first five rows.

Result

The notebook displays the first five employee records.

Business Interpretation

This allows the analyst to quickly understand what the data looks like before performing detailed analysis.

11.2 `tail()`

Syntax

```
df.tail()
```

Purpose

Displays the last five rows.

Business Interpretation

It helps verify that the dataset extends to the expected final records. The notebook displays records ending at index 1469.

11.3 `shape`

Syntax

```
df.shape
```

Result

```
(1470, 35)
```

This means:

- **1,470 rows**
- **35 columns**

11.4 `columns`

Syntax

```
df.columns
```

Purpose

Returns the names of all columns.

Business Interpretation

It provides an overview of the information available for analysis.

11.5 `dtypes`

Syntax

```
df.dtypes
```

Result

The dataset contains:

- 26 `int64` columns
- 9 `object` columns

Business Interpretation

The dataset mainly contains numerical variables along with categorical/text fields.

11.6 `info()`

Syntax

```
df.info()
```

Important Results

- 1,470 entries
- 35 columns
- 26 integer columns
- 9 object columns
- 1,470 non-null values for every field

Business Interpretation

The initial structure of the dataset is consistent and complete with respect to null values.

11.7 `describe()`

Syntax

```
df.describe()
```

The notebook reports, among other values:

Variable	Mean	Minimum	Maximum
Age	36.92	18	60
DailyRate	802.49	102	1499
DistanceFromHome	9.19	1	29
Education	2.91	1	5
EnvironmentSatisfaction	2.72	1	4
HourlyRate	65.89	30	100
JobInvolvement	2.73	1	4
JobLevel	2.06	1	5

Variable	Mean	Minimum	Maximum
TotalWorkingYears	11.28	0	40
TrainingTimesLastYear	2.80	0	6
WorkLifeBalance	2.76	1	4
YearsAtCompany	7.01	0	40
YearsInCurrentRole	4.23	0	18
YearsSinceLastPromotion	2.19	0	15
YearsWithCurrManager	4.12	0	17

The notebook's descriptive statistics show that employee age ranges from 18 to 60, while total working years ranges from 0 to 40.

---

12. Data Cleaning

12.1 Missing-Value Check

The notebook uses:

```
df.isnull()
```

and:

```
df.isnull().sum()
```

The result is **0 missing values in every column**.

Business Impact

No missing-value treatment is required for the currently loaded dataset.

12.2 `fillna()`

The notebook includes:

```
df.fillna(5)
```

However, because the dataset has no missing values, this operation does not have a practical effect on the current data.

Important Note

Replacing missing values with an arbitrary value such as `5` should generally not be done without understanding the meaning and scale of the column.

12.3 `dropna()`

The notebook also includes:

```
df.dropna()
```

Since no missing values are reported, there are no rows to remove because of null values.

12.4 Duplicate Rows

The notebook does **not** show a duplicate-row check such as:

```
df.duplicated().sum()
```

Therefore, the presence or absence of duplicate rows cannot be claimed from the uploaded notebook.

Recommended Check

```
df.duplicated().sum()
```

12.5 Incorrect Data Types

The `info()` and `dtypes` outputs show consistent `int64` and `object` data types.

No date fields are present in the listed columns.

12.6 Extra Spaces and Inconsistent Values

The notebook does not explicitly check for:

- Leading/trailing spaces
- Inconsistent categorical spelling
- Empty strings

Therefore, these issues cannot be confirmed from the current notebook.

12.7 Unnecessary Columns

The notebook does not perform column removal.

However, fields such as `EmployeeCount` and `StandardHours` have constant values in the displayed descriptive statistics:

- `EmployeeCount` = 1
- `StandardHours` = 80

Their standard deviations are 0.

Such constant fields normally provide little analytical variation and can be considered for removal in a modeling-oriented version of the project.

---

13. Exploratory Data Analysis

13.1 Univariate Analysis

Univariate analysis studies one variable at a time.

Age

The dataset reports:

- Mean age: **36.92**
- Median age: **36**
- Minimum age: **18**
- Maximum age: **60**
- Standard deviation: **9.14**

Interpretation

The employee records cover a broad adult age range, with the average age being approximately 37 years.

13.2 Age Distribution

The notebook creates a histogram:

```
sns.histplot(df['Age'], bins=30, kde=True)
```

Chart Title

Distribution of Age

Objective

To understand how employee ages are distributed.

Why Histogram?

A histogram is suitable for showing the frequency distribution of a numerical variable.

Observation

The notebook creates the age distribution visualization, but the stored notebook does not provide a numerical summary of the histogram's individual bins.

Therefore, no unsupported statement about the most common age group is made here.

---

14. Bivariate Analysis

Bivariate analysis studies the relationship between two variables.

14.1 Years at Company vs Monthly Income

The notebook creates:

```
sns.scatterplot(
    x='YearsAtCompany',
    y='MonthlyIncome',
    data=df
)
```

Chart Title

Monthly Income vs. Years at Company

Objective

To examine whether monthly income changes with years spent at the company.

Why Scatter Plot?

A scatter plot is suitable for comparing two numerical variables.

Business Interpretation

This visualization can help HR analysts investigate whether employee tenure is associated with income levels.

The notebook does not provide a calculated regression coefficient or statistical significance test, so a causal relationship should not be claimed.

14.2 Years at Company vs Monthly Income by Attrition

The notebook creates:

```
sns.scatterplot(
    x='YearsAtCompany',
    y='MonthlyIncome',
    hue='Attrition',
    data=df,
    s=100
)
```

Chart Title

Monthly Income vs. Years at Company by Attrition

Objective

To compare the relationship between tenure and income while distinguishing employees according to attrition status.

Why This Chart?

Using color/hue for `Attrition` makes it possible to visually compare employees who left with employees who stayed.

Business Interpretation

This visualization provides a starting point for investigating whether income and tenure patterns differ between attrition groups.

The notebook does not calculate group-level statistics, so the analysis should not claim that one attrition group definitively has higher income or longer tenure based only on this chart.

---

15. Multivariate Analysis

15.1 Correlation Matrix

The notebook creates a correlation heatmap using:

```
df.corr(numeric_only=True)
```

and:

```
sns.heatmap(
    df.corr(numeric_only=True),
    annot=True,
    fmt='.2f',
    cmap='coolwarm',
    linewidths=.5
)
```

Chart Title

Correlation Matrix of Numerical Features

Objective

To examine relationships among numerical variables.

Why Heatmap?

A heatmap makes it easier to visually compare many correlation values at the same time.

Business Interpretation

Correlation can help identify variables that move together.

However:

> **Correlation does not prove causation.**

For example, if two HR variables are strongly correlated, it does not automatically mean that one causes the other.

---

16. Feature Engineering

The current notebook does not create new calculated columns.

Therefore, no completed feature-engineering result should be reported as part of the current analysis.

Recommended Future Features

The following features could be created in an expanded project:

1. Age Group

Example:

```
18–25
26–35
36–45
46–55
56+
```

Business benefit: Makes age-group comparisons easier.

2. Experience Group

Group `TotalWorkingYears` into experience bands.

Business benefit: Helps compare attrition patterns by career experience.

3. Tenure Group

Group `YearsAtCompany` into categories.

Business benefit: Helps HR identify early-tenure and long-tenure employees for further investigation.

4. Income Group

Create low, medium, and high income categories.

Business benefit: Makes income-based comparisons easier.

These are proposed transformations and are **not existing results from the uploaded notebook**.

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17. Data Visualization

The uploaded notebook contains four major visualizations.

Chart	Variables	Purpose
Age Histogram	Age	Understand age distribution
Scatter Plot	YearsAtCompany, MonthlyIncome	Study tenure-income relationship
Attrition Scatter Plot	YearsAtCompany, MonthlyIncome, Attrition	Compare tenure-income patterns by attrition
Correlation Heatmap	Numerical variables	Examine numerical relationships

Visualization Principles

Charts should:

- Have clear titles
- Use meaningful axis labels
- Show relevant variables
- Avoid unnecessary decoration
- Support a specific analytical question

---

18. Business Insights

The following insights are directly supported by the notebook outputs:

Insight 1: Dataset Size

The analysis contains **1,470 employee records and 35 variables**.

Insight 2: No Reported Missing Values

All 1,470 records are non-null across the 35 fields according to `info()`, and `isnull().sum()` reports zero for every column.

Insight 3: Employee Age Range

Employee ages range from **18 to 60 years**, with an average age of approximately **36.92 years**.

Insight 4: Employee Experience

Total working years range from **0 to 40**, with an average of approximately **11.28 years**.

Insight 5: Company Tenure

Years at the company range from **0 to 40 years**, with an average of approximately **7.01 years**.

Insight 6: Income and Tenure Were Explicitly Investigated

The notebook specifically analyzes monthly income against years at the company and then adds attrition as a comparison dimension.

Insight 7: Constant Fields

`EmployeeCount` and `StandardHours` have zero standard deviation in the descriptive statistics, indicating no variation in these fields within the dataset.

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19. Recommendations

Recommendation 1: Analyze Attrition by Department

Create an attrition percentage for each department.

Why: This would help HR determine whether turnover is concentrated in particular organizational areas.

Expected benefit: Better targeting of employee-retention programs.

Recommendation 2: Analyze Attrition by Job Role

Compare attrition across `JobRole`.

Why: Different roles may have different work conditions and employee experiences.

Expected benefit: More focused HR interventions.

Recommendation 3: Investigate Overtime

Analyze `OverTime` against `Attrition`.

Why: Overtime is an available employee-work characteristic and can be directly compared with attrition.

Expected benefit: Helps HR investigate whether workload patterns are associated with employee turnover.

Recommendation 4: Analyze Job Satisfaction

Compare `JobSatisfaction` with `Attrition`.

Why: Job satisfaction is available in the dataset and may provide useful information about employee experience.

Expected benefit: Can help HR identify areas requiring further investigation.

Recommendation 5: Create an HR Dashboard

The analyzed data can be converted into a Power BI dashboard containing:

- Total employees
- Attrition count
- Attrition percentage
- Department-wise attrition
- Job-role-wise attrition
- Overtime analysis
- Job satisfaction
- Monthly income
- Age groups
- Years at company

---

20. Challenges Faced

Challenge 1: Large Number of Variables

The dataset contains 35 columns, making it important to select meaningful variables for visualization.

Challenge 2: Mixed Data Types

The dataset contains both numerical and object/categorical columns.

Challenge 3: Categorical Coding

Several satisfaction and rating fields are represented numerically. Their exact category meanings are not documented inside the uploaded notebook.

Challenge 4: Limited Completed EDA

The notebook performs basic exploration and visualization but does not include completed group-by analysis for all potential HR questions.

Challenge 5: Duplicate Validation

A duplicate-row check is not included in the notebook, so duplicate status cannot be confirmed from the submitted analysis.

---

21. Conclusion

This project performed an exploratory analysis of an HR Employee Attrition dataset using Python.

The dataset contains **1,470 employee records and 35 columns**. The analysis used Pandas, NumPy, Matplotlib, and Seaborn.

The notebook successfully performed:

- Dataset loading
- DataFrame creation
- Initial data inspection
- Shape analysis
- Column inspection

- Data-type inspection
- Missing-value checking
- Descriptive statistics
- Age distribution analysis
- Income and tenure analysis
- Attrition-based visualization
- Correlation analysis

The dataset has no reported missing values, and the numerical variables provide useful information about employee demographics, income, experience, satisfaction, and tenure.

The current project is a good foundation for a more detailed HR analytics solution. The next stage should focus on calculating attrition rates by department, job role, overtime, satisfaction, business travel, and other relevant employee characteristics.

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22. Future Scope

The project can be extended in several ways.

22.1 Power BI Dashboard

Create an interactive dashboard with:

- Attrition KPIs
- Department filters
- Job role filters
- Age analysis
- Income analysis
- Overtime analysis
- Satisfaction analysis

22.2 Predictive Analytics

A machine-learning model could be developed to predict the probability of employee attrition.

Possible algorithms include:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting

22.3 Employee Risk Segmentation

Employees could potentially be grouped into different attrition-risk categories after developing and validating a predictive model.

22.4 Automation

The analysis could be automated so that new HR data is processed regularly.

22.5 Real-Time HR Reporting

A future solution could connect HR data to a dashboard and provide updated workforce metrics.

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23. References

Dataset

The uploaded notebook references:

`WA_Fn-UseC_-HR-Employee-Attrition.csv`

The dataset is loaded directly in the notebook using Pandas.

Python

Python was used as the main programming language for the analysis.

Pandas

Used for:

- Reading CSV data
- DataFrame creation
- Data inspection
- Missing-value analysis
- Descriptive statistics

NumPy

Used as the numerical computing library imported in the notebook.

Matplotlib

Used for plotting and chart creation.

Seaborn

Used for:

- Histograms
- Scatter plots
- Correlation heatmaps

The notebook explicitly imports all four Python analysis libraries:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

Final Project Summary

Area	Result
Dataset	HR Employee Attrition
Records	1,470
Columns	35
Numerical columns	26
Object columns	9
Missing values reported	0
Average employee age	36.92 years
Average total working years	11.28 years
Average years at company	7.01 years
Main analysis type	Exploratory Data Analysis
Visualization tools	Matplotlib + Seaborn
Main subject	Employee attrition

Area	Result
Future extension	Power BI + Predictive Analytics

Overall Outcome: The project establishes a clean exploratory foundation for HR employee-attrition analysis. The strongest next step is to extend the notebook with grouped attrition calculations, feature engineering, additional visualizations, and an interactive HR dashboard.